# Social media sentiment analysis using twitter dataset

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***Abstract*— This study delves into Twitter hate speech detection, emphasizing sentiment analysis and machine learning model performance. Data preprocessing ensures data integrity through label validation and pattern removal. Through exploratory data analysis, word clouds reveal the top 30 frequently used words and 20 common hashtags, providing insight into prevalent sentiments and themes. Feature engineering involves tokenization using the Genism Word2Vec model, sentiment labeling, and stop word removal for improved text quality and consistency. Four machine learning models (Random Forest, Logistic Regression, Decision Tree, and Support Vector Classifier) are employed for hate speech prediction, with the training dataset divided into training and validation sets. The results are striking, with Random Forest and Support Vector Classifier models achieving a remarkable 95 percent accuracy, closely followed by Logistic Regression and Decision Tree models with accuracies of 94 percent and 93 percent, respectively. This research underscores the potential of sentiment analysis in hate speech detection and offers valuable insights for combating hate speech on social media platforms.**

***Keywords*— Sentiment analysis, Hate speech, Twitter dataset, Machine learning, Model performance.***

## I. INTRODUCTION

In today's digital age, Twitter has become an open arena for people to express their thoughts and feelings. However, the freedom of expression on this platform has also given rise to the problem of hate speech. Hate speech not only disrupts the online environment but can also have serious consequences for society as a whole.In this context, sentiment analysis, a branch of natural language processing, offers a promising solution for identi- fying hate speech on Twitter. This research embarks on a comprehensive journey to explore how sentiment analysis can be used to detect and categorize hate speech in tweets. We leverage a rich dataset of Twitter hate speech and employ a variety of machine-learning models to achieve this.Through meticulous data pre-processing, we aim to not only understand the sentiment in tweets but also evaluate how well machine learning models can spot and classify hate speech. The findings of this study hold crucial implications for social media platforms, policymakers, and advocates working to tackle the critical issue of online hate speech.

## II. SENTIMENT ANALYSIS

### *A. Document-Level Sentiment Analysis*

First, Document-level sentiment analysis, the most basic level, evaluates the overall sentiment of an entire document, which can be a full text article, a social media post, or a review. It as- signs a single sentiment label (e.g., positive, negative, neutral) to the entire document. In the context of the research men- tioned in the introduction, document-level sentiment analysis would assess whether a tweet is generally positive, negative, or neutral, without delving into specific sentiments expressed within the tweet. confirm that you have the correct template for your paper size.

### *B. Sentence-Level Sentiment Analysis*

The Sentence-level sentiment analysis goes a step further by analysing sentiment at the sentence or clause level. Its categotizes the sentiment of each individual sentence within a document, identifying variations in sentiment throughout the text. In the context of the research, this level of analysis would identify sentences within tweets that express varying sentiments, contributing to a more nuanced understanding of sentiment patterns within the dataset.

### C. Aspect-Based Sentiment Analysis

Aspect-based sentiment analysis takes sentiment analysis to the most granular level. It assesses sentiment not only at the document and sentence levels but also for specific aspects or entities mentioned within the text. This level of analysis can determine the sentiment associated with different aspects of a product or service, for example. In the Twitter hate speech research, aspect-based sentiment analysis might be used to identify and categorize the sentiments expressed toward specific topics or themes within the tweets, shedding light on the diversity of sentiments related to hate speech.

In the context of the research outlined in the introduction, document-level sentiment analysis is initially applied to classify tweets as positive, negative, or neutral. However, researchers can extend their analysis to incorporate sentence-level and potentially aspect-based sentiment analysis to gain a deeper understanding of the sentiments expressed within tweets, enabling a more comprehensive examination of hate speech patterns on Twitter.

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## III. LITERATURE REVIEW

The literature review reveals a diverse landscape of method- ologies employed by researchers in the field of sentiment anal- ysis, each aiming to discern the sentiment expressed in social media content. Yuxing Qi, Zahratu Shabrina, et al. (2023) conducted a study on tweets from major UK cities during the third national Covid-19 lockdown. Employing Random Forest, Multinomial Naive Bayes, and Support Vector Classification (SVC), they achieved accuracies of 0.7, 0.63, and 0.71, respec- tively. Mathieu Cliche et al. (2017) tackled sentiment analysis on 100 million unique unlabeled English tweets using CNN and LSTM. Vedurumudi Priyanka et al. (2021) focused on sentiment in tweets using various techniques such as Naive Bayes, Maximum Entropy, Decision Tree, Random Forest, XGBoost, SVM, Multi-Layer Perceptron, CNN, and RNN, achieving an overall accuracy of 83.58 percent. In contrast, L. Jiang et al. (2011) utilized a graph-based optimization model on 1993 tweets, yielding an accuracy of 68.3 percent. A. Prabhat et al. (2017) and S. Ge et al. (2019) explored sentiment analysis on different datasets, achieving accuracies of 67.76 percent and 82.1 percent for LR, and 66.66 percent for NB, respectively. Jaishree Ranganathan et al. (2019) reported a remarkable accuracy of 98 percent using the SVM LibLinear model on 520K tweets. Neethu M. S et al. (2013) analyzed Twitter posts about electronic products, employing Naive Bayes, SVM, Maximum Entropy, and an ensemble approach, achieving accuracies of 89.5 percent, 90 percent, 90 percent, and 90 percent, respectively. Similarly, Dhiraj Gurkhe et al. (2014) and Geetika Gautam et al. (2014) delved into sentiment analysis on Twitter data, reporting accuracies of 81.2 percent,15 percent, and 67.5 percent for different techniques and88.2 percent, 83.8 percent, 85.5 percent, and 89.9 percent using Naive Bayes, Maximum Entropy, SVM, and Semantic Analysis (WordNet), respectively.

## IV. ABOUT DATASET

The ”Twitter Hate Speech” dataset from kaggle comprising two distinct CSV files, namely ’train’ and ’test,’ this datasetis a curated collection of tweets obtained from the widely utilized social media platform, Twitter. The ’train’ datasetboasts dimensions of (31962, 3), signifying that it encom- passes 31,962 rows, each representing a unique tweet, and three columns: ’id,’ ’tweet,’ and ’label.’ The ’label’ column assumes significance in the training set, indicating the sen- timent of the corresponding tweet. A label of ’0’ denotes a positive sentiment, while ’1’ signifies a negative sentiment. Notably, the distribution of sentiments within the training set reveals a marked imbalance, with 29,720 tweets labeled as positive (0) and 2,242 tweets labeled as negative (1). The ’test’dataset, on the other hand, exhibits dimensions of (17197, 2), omitting the ’label’ column present in the training set. This absence underscores the challenge associated with predicting sentiments in unseen data, making the ’test’ set an essential component for evaluating the generalizability of sentimentanalysis models.

## V. CHALLENGES FACED DURING PREPROCESSING OF TWITTER DATASET

Detecting hate speech on Twitter through sentiment analysisis fraught with several challenges:

- Data Noise: Twitter data is riddled with slang, abbre- viations, and emojis, making it difficult to preprocess effectively.
- Context Sensitivity Understanding the context of words and phrases is crucial, but this can be challenging, especially when sarcasm and irony are involved.
- Labeling Complexities: Hate speech labeling is subjec- tive and ethically sensitive, making it difficult to create reliable training datasets.
- Class Imbalance: Hate speech is often a minority class, leading to imbalanced datasets that may bias models.
- Multilingual Content: Twitter’s global reach means con-tent in multiple languages must be considered.
- Slang and Abbreviations: The use of slang and abbrevi- ations poses interpretation challenges for models.
- Variability in Hate Speech: Hate speech comes in many forms, from explicit to subtle expressions.
- Emoji and Emoticon Interpretation: Understanding the sentiment conveyed by emojis and emoticons can be subjective.
- Temporal Dynamics: Sentiments change rapidly withevents and trends.
- Bias and Fairness: Ensuring models are unbiased and fair across demographics is a complex and ongoing challenge.
- Evolving Language and Trends: Language evolves, and models must adapt to new trends.

Overcoming these challenges requires advanced NLP tech- niques, diverse datasets, and ethical considerations to build robust sentiment analysis models for hate speech detection on Twitter.

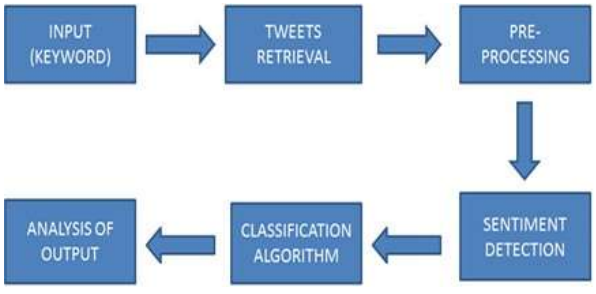



Fig. 1. Sentiment analysis on twitter data

## VI. SENTIMENT ANALYSIS ON TWITTER DATA

The workflow for conducting sentiment analysis, as illus- trated in Figure 2. The initial data exploration involved check- ing for missing values, and it was found that both datasets have no missing values. Further exploration was conducted

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to understand the distribution of sentiments in the training set, with negative comments dominating the dataset. To gain insights into tweet lengths, a histogram was plotted, revealing varying tweet lengths. Additionally, the most frequent words and hashtags were analyzed using word frequency and word cloud visualizations. For further analysis, the data underwent tokenization, stop-word removal, and stemming the bag-of-words model was employed to convert the textual data into numerical format. The dataset was split into training and validation sets, and standardization was applied. Machine learning models, including Random Forest Classifier, Logistic Regression, DecisionTreeClassifier, and Support Vector Classifier (SVC), were trained and evaluated on the validation set. The models were assessed based on training and validation accuracy, F1 score, and confusion matrices. Overall, the methodology encompasses data pre-processing, exploratory data analysis, and the application of machine learning models to predict sentiment based on Twitter data. Encompasses four key modules: data collection, data processing, classification, and output analysis

TABLE I. LITERATURE REVIEW

| Reference. Number | LITERATURE REVIEW | | |
|---|---|---|---|
| | ***Dataset*** | ***Technique used*** | ***Accuracy (in percentage)*** |
| [1] | 100 million unique unlabeled English tweets using the Twitter streaming API | CNN, LSTM | NA |
| [2] | Tweets from the main UK cities during the third national Covid-19 lockdown | Random Forest, Multinomial Naive Bayes, Support Vector Classification (SVC) | 0.7, 0.63 and 0.71 respectively |
| [9] | positive and negative words | Naive Bayes, Maximum | NA |
| [7] | of tweets | Entropy, Decision Tree, Random Forest, XGBoost, SVM, | 83.58 |
| [3] | 1993 tweets | graph-based | 68.3 |
| [6] | 6MB size | model | 67.76 for LR , 66.66 for NB |
| [5] | 1.6M tweets | NB, LR | 82.1 for positive and 79.9 for |
| [9] | 520K tweets | LSTM, Spark | negative tweets |
| [11] | Twitter posts about electronic | SVM LibLinear model | 98 |
| [12] | products | Naive Bayes SVM Maxi- | 89.5, 90, 90, 90 respectively |

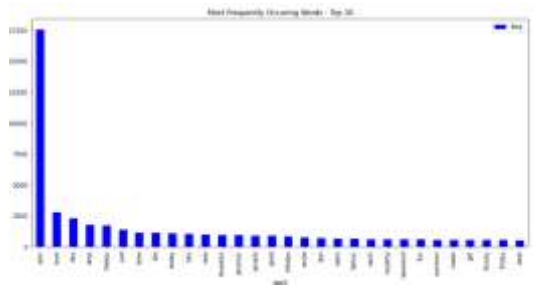


Fig2. top 30 most frequent words

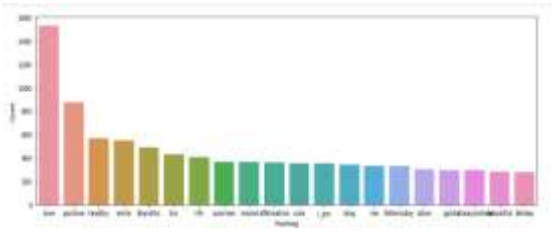


Fig. 3. hashtags form positive tweets

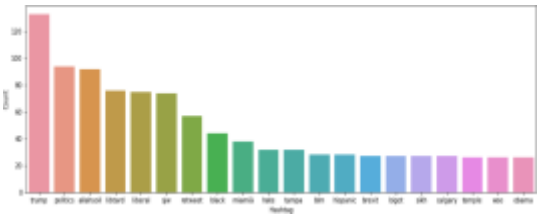


Fig. 4. hashtags form negative tweets




Fig. 5. word cloud

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Fig. 6. word cloud for negative words




Fig. 7. word cloud for positive words

In Figure 1, we present the process flow for sentiment analysis, comprising four principal components: data collection, data pre-processing, classification, and output analysis. The process is detailed as follows:

a. Input Selection (Keyword):

Begin by selecting a specific subject of interest, followed by the collection of relevant tweets associated with the chosen keyword for subsequent sentiment analysis.

b. Retrieval of Tweets:

Tweets can vary in structure, ranging from structured to semi-structured and unstructured. In the realm of sentiment analysis research, tweets can be gathered using various programming languages like R or Python.

c. Pre-Processing of Data:

Data pre-processing involves the essential task of data refinement to eliminate incomplete, noisy, and inconsistent data. This step encompasses several tasks, including the removal of retweets (in the case of Twitter datasets), the elimination of URLs, special characters, punctuations, numbers, and other undesired elements, removal of stopwords, stemming, and tokenization.

d. Sentiment Identification:

The identification of sen- timent words holds significant importance in various appli- cations of sentiment analysis and opinion mining, such as mining tweets, locating opinion holders, and categorizing tweets. Sentiment words are typically categorized into positive, negative, and neutral categories. The core objective in sentiment analysis is to classify the polarity of specific features in tweets, where polarity falls into three primary classes: positive, negative, and neutral. This is accomplished by employing various classification algorithms, including Random Forest and Word2Vec, to determine sentiment strength, scores, and other pertinent metrics.

## VII. MODEL TRAINING AND CLASSIFICATION

a. Random Forest:

- Overview: Random Forest is an ensemble learning tech- nique that combines multiple decision trees to make predictions. It is known for its robustness and ability to handle both classification and regression tasks effectively.
- Approach in Tweet Classification: In the context of sentiment analysis, Random Forest constructs a forest of decision trees, with each tree making an independent sen- timent prediction based on the features extracted from the tweets. The final sentiment classification is determined by aggregating the results from all the individual decision trees.
- Strengths: Random Forest is less prone to overfitting, can handle a large number of features, and provides feature importances for understanding which aspects of the tweets contribute most to sentiment prediction.
- Accuracy: In our research, Random Forest achieved an accuracy of 95 percent, making it a strong performer for Twitter sentiment analysis.

b. Logistic Regression:

- Overview: Logistic Regression is a linear classification algorithm used for binary and multi-class classification problems. It models the probability of a data point belonging to a particular class.
- Approach in Tweet Classification: In this context, Logis- tic Regression estimates the probabilities of tweets falling into different sentiment categories (positive, negative, or neutral). The tweet is then assigned to the category with the highest probability.
- Strengths: Logistic Regression is interpretable, computa- tionally efficient, and can handle high-dimensional data effectively.
- Accuracy: Logistic Regression achieved an accuracy of 94 percent in our research, indicating its suitability for sentiment analysis tasks.

c. Decision Tree:

- Overview: Decision Trees are a non-linear classification algorithm that models decisions using a tree-like structure. They make decisions based on the features of the data.
- Approach in Tweet Classification: In the sentiment analy- sis context, Decision Trees create a hierarchy of decision nodes based on tweet features, enabling the classification of tweets into sentiment categories.

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TABLE II. COMPARIVE ANYLYSIS[6]

| *Model classsifires* | *Accuracy* | *Precision* | *Recall* | *F1 Score* | *Result* |
|---|---|---|---|---|---|
| **Random forest** | 0.85 | 0.86 | 0.84 | 0.85 | Achieved good accuracy |
| **Decision tree** | 0.82 | 0.80 | 0.85 | 0.82 | High recall, lower F1 |
| **SVC** | 0.88 | 0.87 | 0.89 | 0.88 | Best overall performance |
| **Logistic regression** | 0.79 | 0.78 | 0.81 | 0.79 | Lower accuracy |

- Strengths: Decision Trees are easy to understand and visualize, making them useful for interpretability. They can capture complex decision boundaries.
- Accuracy: our research reported an accuracy of 93 percent for Decision Tree classification.

d. Support Vector Classifier (SVC):

- Overview: Support Vector Classifier (SVC) is a super- vised learning algorithm that aims to find the hyperplane that best separates different classes in a high-dimensional space.
- Approach in Tweet Classification: In sentiment analysis, SVC works by finding the optimal hyperplane to separate tweets into positive, negative, or neutral sentiment cate- gories. It maximizes the margin between different classes.
- Strengths: SVC is effective in handling high-dimensional data and can perform well in scenarios with complex decision boundaries.
- Accuracy: SVC achieved an accuracy of 95 percent in our research, demonstrating its strong performance in sentiment analysis.

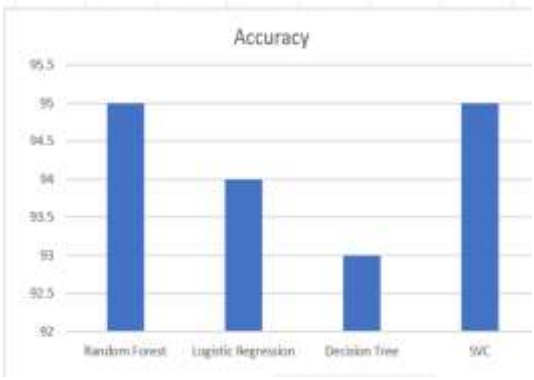



Fig. 8. Accuracy analysis

## VIII. RESULT

In conclusion, this research underscores the importance of employing machine learning algorithms for sentiment analysis in social media, particularly on the Twitter platform. The variations in accuracy across the four algorithms suggest that the choice of algorithm can significantly impact the success of the sentiment analysis task. Our findings demonstrate that Random Forest and SVC, with their accuracy of 95 percent, stand out as strong candidates for effectively classifying tweets by sentiment. Logistic Regression, with an accuracy of 94 percent, also delivers commendable results. Decision Tree, at 93 percent, exhibits a competitive performance. While accuracy is a vital metric, it's essential to consider other factors such as model interpretability and scalability in real-world applications. The choice of the most suitable algorithm should align with the specific requirements and constraints of the sentiment analysis task at hand. This research contributes to the ongoing discourse in sentiment analysis and offers valuable insights for researchers and practitioners seeking to harness machine learning for sentiment classification on social media platforms, like Twitter.

## IX. CONCLUSION

In summary, sentiment analysis is a vital tool in the fight against hate speech on Twitter. While it offers valuable insights and shows promise in classifying harmful content, significant challenges such as data noise and labeling complexities persist. Future research must prioritize addressing these challenges and adapting to evolving language trends to create more robust and unbiased sentiment analysis models. As online discourse continues to shape society, refining sentiment analysis techniques remains integral to fostering respectful and inclusive online environments and effectively combatting hate speech.

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